

Software Requirements Specification

for

Electronics Store Inventory Management System

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version
Team	03/09/2026	Initial SRS preparation	1.0
Team	07/09/2026	Aligned SRS with simplified current-quantity scope and updated use-case model	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for the Electronics Store Inventory Management System (ESIMS), Version 1.0. It describes the system scope, intended users, interfaces, system features, quality requirements, business rules, data fields, and requirement traceability information.

1.2 Intended Audience and Reading Suggestions

- Developers: use the requirements as the baseline for system design and implementation.
- Testers: derive unit, integration, and system test cases from the functional requirements.
- Project faculty/manager: evaluate project scope, feature ownership, and progress.
- Store users/stakeholders: understand the expected behavior and capabilities of the system.

The document is organized according to the supplied SRS template: introduction, overall description, external interfaces, analysis models, system features, nonfunctional requirements, other requirements, and appendices.

1.3 Product Scope

ESIMS is a web-based application for maintaining the current inventory of products in a small-to-medium electronics retail store. It replaces manual or spreadsheet-based tracking with a centralized product database. Users can authenticate, view and search products, add or delete products, update current quantities/details, and obtain availability information. The system automatically classifies products as In Stock, Low Stock, or Out of Stock using quantity and reorder-level rules, and provides an availability report and dashboard.

Scope clarification: Version 1.0 maintains only the current available quantity for each product. It does not maintain Stock In/Stock Out transactions, previous quantities, transaction history, or transaction audit records.

1.4 References

- Electronics Store Inventory Management System — Group 16 Project Proposal.
- Finalized functional and nonfunctional requirements for ESIMS.
- SRS structure/template supplied for the Software Engineering project.

2. Overall Description

2.1 Product Perspective

ESIMS is a new, self-contained academic software system. The proposed architecture consists of a browser-based front end, an application/backend layer, and a relational database that stores user and inventory information. The system does not depend on external e-commerce, payment, barcode, or supplier-integration systems in Version 1.0.

2.2 Product Functions

- Authenticate users and enforce role-based authorization.
- View the current inventory.
- Search products by product ID, name, or category.
- Add new product records.
- Delete existing product records with confirmation.
- Update current quantity or other permitted product details.
- Validate product information and prevent duplicate product IDs.
- Automatically classify availability as In Stock, Low Stock, or Out of Stock.
- Generate and filter an availability report.
- Display a dashboard with current inventory summary metrics.
- Allow administrators to manage users and system settings.

2.3 User Classes and Characteristics

User Class	Characteristics	Typical Access
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Administrator	Has the highest privilege level and performs administrative functions.	User management, system settings, inventory administration, reports and dashboard.
Store Employee	Routine operational user with basic computer skills.	Login, view/search inventory, and permitted inventory operations.
Store Manager	Frequent inventory/management user responsible for monitoring stock.	Inventory monitoring, update operations, availability report and dashboard.
Management	Primarily interested in summarized inventory information.	Dashboard and availability reports, normally read-only.

2.4 Operating Environment

- Client: desktop/laptop computer with a current web browser such as Google Chrome, Mozilla Firefox, or MicrosoftEdge.
- Server: standard application/web server suitable for an academic web application.
- Database: relational SQL database such as MySQL or PostgreSQL.

2.5 Design and Implementation Constraints

- Core inventory data shall be stored in a relational SQL database.
- The system shall be implementable on standard laboratory/personal computer hardware.
- The project must be completed within the academic semester timeline.
- The implementation shall use browser-based HTTP/HTTPS communication.

2.6 Assumptions and Dependencies

- Users have valid accounts and appropriate permissions.
- Inventory data entered by authorized users is reasonably accurate.
- The application server and database are available during normal operation.
- No payment, e-commerce, barcode, or supplier integration is required for Version 1.0.

3. External Interface Requirements

3.1 User Interfaces

- Login screen with username/email and password fields.
- Inventory screen showing product ID, name, category, price, current quantity, reorder level, and availability status.
- Search/filter controls for product ID, product name, and category.
- Add-product form with validation and duplicate-ID checking.
- Delete action with a confirmation prompt.
- Update-inventory form for current quantity and permitted product details.
- Availability report screen with status filtering.
- Dashboard containing current inventory summary metrics.
- Clear validation and error messages shall be displayed near the relevant action or field.

3.2 Software Interfaces

- A relational database shall provide persistent storage for user and inventory data.
- The application/backend shall communicate with the database using a suitable database connectivity mechanism.
- The browser client shall communicate with the application over HTTP/HTTPS.
- Authentication/session handling shall control access to protected functions.

3.3 Communications Interfaces

Client-server communication shall use standard HTTP/HTTPS. HTTPS shall be preferred for deployed operation so that credentials and inventory information are protected during transmission. No external messaging, email, or payment communication is required in Version 1.0.

4. Analysis Models

The primary analysis model for this SRS is the UML use-case diagram below. It identifies the major actors, system functions, and include relationships. The diagram has been aligned with the simplified project scope: the system maintains current quantity only and does not model Stock In/Stock Out transaction history.

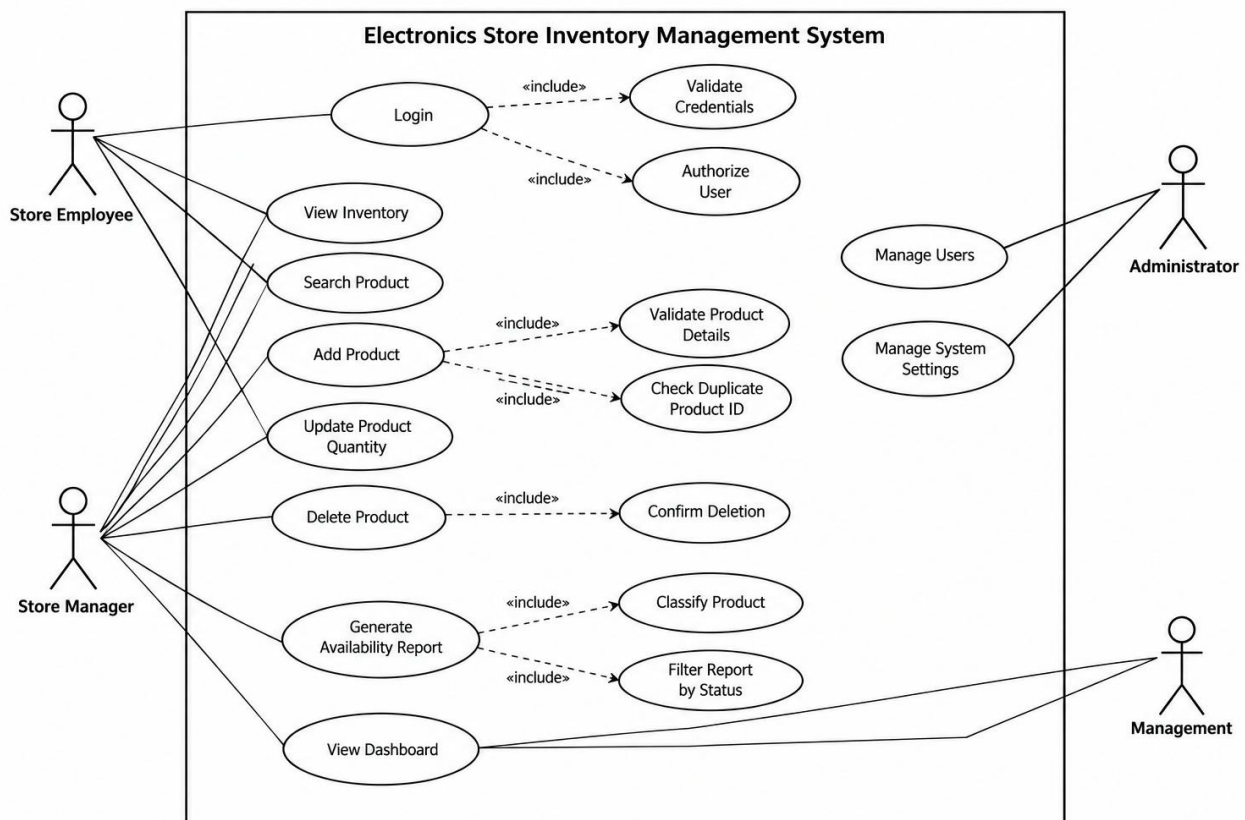


Figure 4.1 — Use Case Diagram for ESIMS

The main actors are Store Employee, Store Manager, Administrator, and Management. The include relationships show mandatory supporting behavior such as product validation, duplicate-ID checking, deletion confirmation, and availability classification.

5. System Features

5.1 User Authentication and Authorization

5.5.1 Description and Priority — High

The system authenticates users before protected functions are accessed and applies role-based permissions.

5.1.2 Stimulus/Response Sequences

- User opens the login page → system displays login fields.
- User submits valid credentials → system authenticates the user and opens the permitted interface.
- User submits invalid/inactive credentials → system denies access and displays an error.
- Authenticated user requests a protected operation → system checks the user's role before allowing the operation.

Functional Requirements

Req ID	Requirement
FR-AUTH-01	The system shall provide a login form requiring a username/email and password.
FR-AUTH-02	The system shall authenticate submitted credentials against stored account information.
FR-AUTH-03	The system shall deny access for invalid or inactive accounts.
FR-AUTH-04	The system shall enforce role-based access to protected functions.
FR-AUTH-05	The system shall provide logout that terminates the active session.
FR-AUTH-06	The system shall not display passwords in plain text in the interface.

5.2 Inventory Management

5.5.1 Description and Priority — High

Authorized users shall be able to view, search, add, delete, and update the current inventory records.

5.2.2 Stimulus/Response Sequences

- User selects View Inventory → system displays current product records.
- User enters search criteria → system returns matching products.
- User enters a new product → system validates fields and duplicate ID, then saves the record.
- User chooses Delete → system requests confirmation and removes the record if authorized.
- User updates quantity/details → system validates the new values and stores the current inventory state.

Functional Requirements

Req ID	Requirement
FR-INV-01	The system shall allow authorized users to add an inventory item.
FR-INV-02	Each inventory item shall have a unique product/item ID.
FR-INV-03	The system shall store product name, category, quantity, unit price, reorder level, and optional location.
FR-INV-04	Authorized users shall be able to update current quantity and permitted product details.
FR-INV-05	The system shall require confirmation before deletion.
FR-INV-06	The system shall prevent deletion by unauthorized users.
FR-INV-07	The system shall display complete stored product details.
FR-INV-08	The system shall support search/filter by product ID, name, and category.
FR-INV-09	Negative quantity, unit price, or reorder level values shall be rejected.

5.3 Availability Classification and Reporting

5.5.1 Description and Priority — High

The system derives a product's availability status from its current quantity and reorder level and provides a report of the current status of products.

5.3.2 Stimulus/Response Sequences

- System reads a product's current quantity and reorder level → system determines the availability status.

- User opens the availability report → system displays current products and statuses.
- User selects a status filter → system displays only matching products.

Functional Requirements

Req ID	Requirement
FR-REP-01	Quantity = 0 shall result in Out of Stock status.
FR-REP-02	Quantity > 0 and quantity ≤ reorder level shall result in Low Stock status.
FR-REP-03	Quantity > reorder level shall result in In Stock status.
FR-REP-04	Availability status shall be shown in inventory views.
FR-REP-05	The system shall generate a report containing product ID, name, category, quantity, reorder level, and status.
FR-REP-06	The availability report shall support filtering by availability status.

5.4 Dashboard

5.5.1 Description and Priority — Medium-High

The dashboard summarizes the current state of inventory using simple metrics for quick decision-making.

5.4.2 Stimulus/Response Sequences

- Authorized user opens the dashboard → system retrieves current inventory data.
- System calculates/display total products and total stock quantity.
- System displays counts of low-stock and out-of-stock products.
- User selects inventory/report navigation → system opens the corresponding screen.

Functional Requirements

Req ID	Requirement
FR-DASH-01	The dashboard shall display the total number of inventory items.
FR-DASH-02	The dashboard shall display the total current stock quantity.
FR-DASH-03	The dashboard shall display the number of low-stock items.
FR-DASH-04	The dashboard shall display the number of out-of-stock items.
FR-DASH-05	The dashboard shall provide navigation to inventory and availability reports.
FR-DASH-06	Dashboard values shall reflect the current stored inventory data.

5.5 Validation and Error Handling

5.5.1 Description and Priority — High

The system shall prevent invalid inventory records and provide understandable feedback when operations fail.

5.5.2 Stimulus/Response Sequences

- User submits an incomplete or invalid form → system identifies the invalid fields and prevents saving.
- User submits an existing product ID → system rejects the record as a duplicate.
- User submits a negative numeric value → system rejects the value.
- Unexpected application/database error occurs → system displays a safe error message without exposing sensitive implementation details.

Functional Requirements

Req ID	Requirement
FR-VAL-01	Mandatory fields shall be validated before saving.
FR-VAL-02	Duplicate product IDs shall be rejected.
FR-VAL-03	Invalid numeric values shall be rejected.
FR-VAL-04	Error messages shall clearly indicate the problem and, where practical, how to correct it.
FR-VAL-05	Server/database errors shall not expose passwords, stack traces, or sensitive implementation details.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- NFR-01: The system shall load the inventory list within a reasonable response time under normal operating conditions.
- NFR-02: The system shall provide a prompt response to add, delete, update, and search actions under normal operating conditions.

6.2 Safety Requirements

NFR-03: No direct physical safety risk is introduced because ESIMS does not control physical equipment. The system shall nevertheless prevent invalid inventory quantities such as negative values to avoid misleading operational decisions.

6.3 Security Requirements

- NFR-04: Only authenticated and authorized users shall be able to perform protected inventory operations.
- NFR-05: Passwords shall not be displayed in plain text in the user interface.
- NFR-06: The system shall enforce role-based authorization for administrative and inventory functions.
- NFR-07: Sensitive application/database errors shall not expose passwords or implementation details.

6.4 Software Quality Attributes

- NFR-08 Usability: The interface shall be simple enough for staff with basic computer literacy to use without formal training.
- NFR-09 Reliability: Inventory changes shall be persisted correctly under normal operating conditions.
- NFR-10 Availability: The application shall be available during expected store operating hours with minimal unplanned downtime.
- NFR-11 Maintainability: The implementation shall be modular so that additional inventory features can be added with limited rework.
- NFR-12 Scalability: The system shall support growth in catalog size and number of users without major redesign.
- NFR-13 Compatibility: The system shall function correctly on current versions of Chrome, Firefox, and Edge.
- NFR-14 Recoverability: The database should support periodic backups and restoration from the latest valid backup.
- NFR-15 Accuracy: Dashboard and report values shall accurately reflect the current database state when generated.

6.5 Business Rules

- BR-01: Every inventory item must have a unique product/item ID.
- BR-02: Inventory quantity cannot be negative.
- BR-03: Quantity = 0 means Out of Stock.
- BR-04: Quantity > 0 and Quantity ≤ Reorder Level means Low Stock.
- BR-05: Quantity > Reorder Level means In Stock.
- BR-06: Only authorized roles may add, update, or delete inventory records.
- BR-07: Mandatory fields must be present before an inventory record is saved.
- BR-08: Availability status is derived from current quantity and reorder level; it is not maintained as stock transaction history.
- BR-09: Version 1.0 does not maintain Stock In/Stock Out transaction records or previous/new quantity history.

7. Other Requirements

7.1 Database Requirements

- The system shall use a relational database for persistent inventory and user data.
- Primary keys shall uniquely identify database records.
- Product/item ID shall have a uniqueness constraint.
- Foreign keys may be used where categories or other normalized entities are represented separately.
- Current quantity, reorder level, and availability status shall be available for inventory processing.
- Creation and update timestamps should be maintained for inventory records.

7.2 Backup and Recovery

- The deployed database should be backed up periodically.
- Restoration should recover inventory data from the latest valid backup.
- A reproducible sample dataset should be maintained for demonstration and testing.

Appendix A: Glossary

Term	Meaning
Inventory	Collection of electronic products maintained by the store.
Product ID	Unique identifier assigned to an inventory product.
Quantity	Current number of units available for a product.
Reorder Level	Threshold at or below which a positive quantity is considered Low Stock.
Availability Status	Derived classification: In Stock, Low Stock, or Out of Stock.
CRUD	Create, Read, Update, and Delete operations.
RBAC	Role-Based Access Control.
Dashboard	Screen presenting summarized inventory statistics.
SRS	Software Requirements Specification.

Appendix B: Field Layouts

B.1 Inventory Item Fields

Field	Length	Data Type	Description	Mandatory
Product ID	30	Alphanumeric	Unique product identifier	Y
Product Name	100	String	Electronic product name	Y
Category	50	String/FK	Product category	Y
Price	10,2	Decimal	Price per unit	Y
Quantity	Integer	Numeric	Current available quantity	Y
Reorder Level	Integer	Numeric	Threshold used for Low Stock classification	Y
Location	50	String	Shelf/bin/store location	N
Status	20	Derived	In Stock / Low Stock / Out of Stock	Y
Created At	DateTime	DateTime	Record creation timestamp	Y
Updated At	DateTime	DateTime	Last modification timestamp	Y

B.2 Availability Report Requirements

Report Field	Description
Product ID	Unique product identifier.
Product Name	Inventory product name.
Category	Product category.
Price	Current unit price.

Quantity	Current available quantity.
Reorder Level	Configured low-stock threshold.
Availability Status	In Stock, Low Stock, or Out of Stock.
Location	Optional inventory location.

B.3 Dashboard Requirements

- Total number of inventory items.
- Total current stock quantity.
- Number of In Stock items.
- Number of Low Stock items.
- Number of Out of Stock items.
- Quick navigation to inventory and availability report.

Appendix C: Requirement Traceability Matrix

The following initial traceability matrix connects selected requirements to the planned architecture, design area, code/module area, and test cases. Code-file references can be finalized after implementation.

Req ID	Brief Description	Architecture	Design	Code/File Reference	Test Case
FR-AUTH-01	Login form	Authentication	Login UI	auth/login	TC-AUTH-01
FR-AUTH-04	Role access	Authorization/RBAC	Access control	auth/rbac	TC-AUTH-04
FR-INV-01	Add product	Inventory	Add form	inventory/create	TC-INV-01
FR-INV-05	Delete confirmation	Inventory	Delete flow	inventory/delete	TC-INV-05
FR-INV-08	Search/filter	Inventory	Inventory list	inventory/search	TC-INV-08
FR-REP-02	Low-stock status	Reports	Classification logic	reports/availability	TC-REP-02
FR-REP-05	Availability report	Reports	Report layout	reports/availability	TC-REP-05
FR-DASH-01	Total item count	Dashboard	Dashboard UI	dashboard	TC-DASH-01
FR-DASH-04	Out-of-stock count	Dashboard	Dashboard KPI	dashboard	TC-DASH-04
FR-VAL-02	Duplicate ID prevention	Validation	Input validation	inventory/validation	TC-VAL-02

Sample Test Cases

Test ID	Scenario	Input/Action	Expected Result
TC-AUTH-01	Valid login	Enter valid username/password and submit.	Dashboard or permitted home screen opens.
TC-AUTH-02	Invalid login	Enter incorrect credentials.	Access denied with clear error.
TC-INV-01	Add product	Enter valid mandatory fields.	New product is stored.
TC-INV-02	Duplicate product ID	Enter an existing product ID.	Record is rejected and duplicate error shown.
TC-INV-03	Update quantity	Change current quantity to a valid non-negative value.	New current quantity is stored and status is recalculated.
TC-REP-01	Out of stock	Quantity = 0.	Status becomes Out of Stock.
TC-REP-02	Low stock	$0 < \text{quantity} \leq \text{reorder level}$.	Status becomes Low Stock.
TC-REP-03	In stock	Quantity > reorder level.	Status becomes In Stock.
TC-DASH-01	Dashboard values	Open dashboard after inventory changes.	KPIs match current database values.